

Name: _____

Date: _____

Introduction to Pythagoras' Theorem

Year 8 — Stage 4 Mathematics

Learning Intentions

- I can identify the hypotenuse in a right-angled triangle.
- I can use Pythagoras' theorem to calculate the hypotenuse.
- I can rearrange Pythagoras' theorem to find a shorter side.
- I can apply Pythagoras' theorem to a real-world situation.

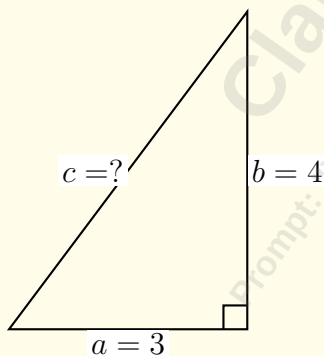
Key Rule

In a right-angled triangle, the **hypotenuse** c (longest side, opposite the right angle) satisfies:

$$c^2 = a^2 + b^2 \implies c = \sqrt{a^2 + b^2}$$

To find a **shorter side**: $a^2 = c^2 - b^2 \implies a = \sqrt{c^2 - b^2}$

Worked Example — Finding the Hypotenuse



Step 1: Write the theorem.

$$c^2 = a^2 + b^2$$

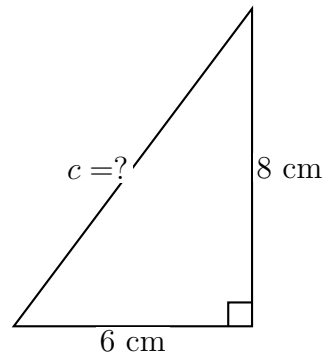
Step 2: Substitute and evaluate.

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

Step 3: Square root both sides.

$$c = \sqrt{25} = 5$$

1. **[Fluency]** Find the length of the hypotenuse c in the triangle below. Show all working.



2. **[Misconception Check]** A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

(a) Find the length of the unknown shorter side a . Show full working.

(b) A student attempted the same problem and wrote:

$$a^2 = 5^2 + 13^2 = 25 + 169 = 194, \quad \text{so} \quad a = \sqrt{194} \approx 13.9 \text{ cm.}$$

Identify the exact error the student made. Then write the correct solution.

3. [Challenge — Real-World Application] A 5 m ladder leans against a vertical wall. The foot of the ladder sits on level ground, 1.8 m from the base of the wall.
- (a) Draw and label a right-angled triangle to represent this situation. Mark the hypotenuse, both known measurements, and the unknown height h .
- (b) Calculate h , the height the ladder reaches up the wall. Give your answer correct to one decimal place.
- (c) The foot of the ladder is then moved to 2.5 m from the wall. The ladder length stays the same.
Without fully recalculating, **explain** whether the ladder now reaches *higher* or *lower* up the wall, and why.

Teacher Answer Key

For teacher use only — not for student distribution

1. Fluency — Finding the hypotenuse (6–8–10 triple)

$$\begin{aligned} c^2 &= 6^2 + 8^2 \\ &= 36 + 64 \\ &= 100 \\ c &= \sqrt{100} = \mathbf{10 \text{ cm}} \end{aligned}$$

Marking: 1 mark — correct substitution into $c^2 = a^2 + b^2$. 1 mark — correct answer $c = 10 \text{ cm}$.

2. Misconception Check — Finding a shorter side (5–12–13 triple)

(a)

$$\begin{aligned} a^2 &= c^2 - b^2 \\ &= 13^2 - 5^2 \\ &= 169 - 25 \\ &= 144 \\ a &= \sqrt{144} = \mathbf{12 \text{ cm}} \end{aligned}$$

Marking: 1 mark — correct rearrangement $a^2 = c^2 - b^2$. 1 mark — correct answer $a = 12 \text{ cm}$.

(b) The student **added** the squares ($5^2 + 13^2$) instead of **subtracting** ($13^2 - 5^2$). When finding a shorter side, the theorem must be rearranged to $a^2 = c^2 - b^2$. A further red flag: the student's answer ($\approx 13.9 \text{ cm}$) is *larger* than the hypotenuse (13 cm), which is geometrically impossible. Correct answer: $a = 12 \text{ cm}$.

Marking: 1 mark — correctly identifies addition error. 1 mark — corrected working and answer.

IMPORTED FROM CHAT:

[TEACHER ONLY] Refinement suggestion — add student-facing mark allocations. Currently no marks appear beside each part. For NSW Stage 4 exam readiness, consider annotating each part with $(__/2)$ or similar so students practise gauging how much to write. Q2(b) in particular could become a 2-mark response question with an explicit instruction to give two distinct points (name the error; explain why the answer is impossible). This also doubles as a useful self-assessment prompt.

Claude Chat: <https://claude.ai/share/84170245-13fd-472f-a116-876fc87809b7>

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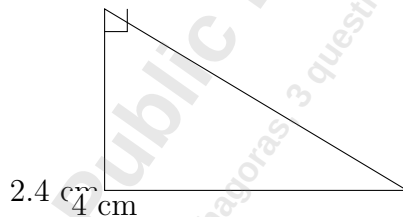
Introduction to Pythagoras' Theorem

Key Idea: In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.

$$c^2 = a^2 + b^2$$

1. A right-angled triangle has legs $a = 3$ cm and $b = 4$ cm. Find the length of the hypotenuse c .

2. The triangle below is right-angled at B . Use the diagram to find the length of the hypotenuse.



3. A student claims that if $a = 5$ and $b = 12$, then the hypotenuse is $c = 17$ because “you just add the sides”. Explain why this is incorrect and calculate the correct value of c .

Teacher Answer Key

4. Solution:

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$c = 5 \text{ cm}$$

Marking guidance: correct substitution; correct square/sum; correct square root.

5. Solution:

$$c^2 = 2.4^2 + 4^2 = 5.76 + 16 = 21.76$$

$$c = \sqrt{21.76} \approx 4.66 \text{ cm}$$

Marking guidance: diagram interpreted correctly; right angle identified; rounding acceptable to two decimal places.

6. Solution:

The claim $c = a + b$ is incorrect because Pythagoras' Theorem uses squares, not addition.

$$c^2 = 5^2 + 12^2 = 25 + 144 = 169$$

$$c = 13$$

Marking guidance: clear explanation of misconception; correct working; correct final value.

IMPORTED FROM CHAT:

— [TEACHER ONLY] — Consider adding one more representation-shift item (e.g., a rotated triangle) to strengthen transfer.

Copilot Chat:

<https://copilot.microsoft.com/shares/FMudwG7axsTVEF3UnXSeY>

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Year 8: Introduction to Pythagoras' Theorem

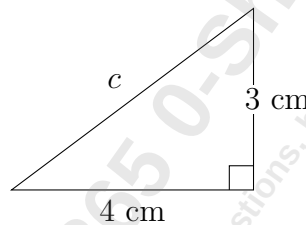
Pythagoras' Theorem applies to right-angled triangles.

$$a^2 + b^2 = c^2$$

where c is the hypotenuse (the longest side).

1. Getting Started

The triangle below is right-angled.



Use Pythagoras' Theorem to find the value of c .

2. Building Confidence

A right-angled triangle has side lengths 5 cm and 12 cm.

Find the hypotenuse.

3. Challenge

The hypotenuse of a right-angled triangle is 13 cm.

One shorter side is 5 cm.

Find the length of the remaining side.

Teacher Answer Key

4.

$$c^2 = 4^2 + 3^2$$

$$c^2 = 16 + 9 = 25$$

$$c = 5 \text{ cm}$$

Marking guidance: Substitutes correctly into the formula and calculates $c = 5 \text{ cm}$.

5.

$$c^2 = 5^2 + 12^2$$

$$c^2 = 25 + 144 = 169$$

$$c = 13 \text{ cm}$$

Marking guidance: Identifies the hypotenuse and applies the theorem correctly.

6.

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = 12 \text{ cm}$$

Marking guidance: Rearranges correctly and solves for the unknown side.

IMPORTED FROM CHAT:

— [TEACHER ONLY] — Add a fourth “spot the error” question where a student incorrectly calculates $3^2 + 4^2 = 7^2$ to explicitly target a common Year 8 misconception about squaring numbers.

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Year 8 Introduction to Pythagoras' Theorem

Key Rule

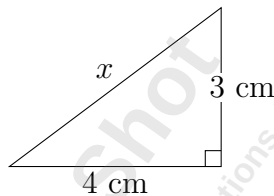
For a right-angled triangle with shorter sides a and b , and longest side c :

$$a^2 + b^2 = c^2$$

The longest side opposite the right angle is called the hypotenuse.

1. Starter

Find the length of the hypotenuse.



Find x .

2. Practice

A right-angled triangle has a hypotenuse of 13 cm and one shorter side of 5 cm.

Find the length of the other side.

3. Challenge

A ladder is leaning against a wall. The foot of the ladder is 6 m from the wall and the ladder itself is 10 m long.

How high up the wall does the ladder reach?

Give your answer in metres.

Teacher Answer Key

1.

$$x^2 = 4^2 + 3^2$$

$$x^2 = 16 + 9$$

$$x^2 = 25$$

$$x = 5$$

Answer: 5 cm

Marking guidance:

- Substitutes correctly into Pythagoras' theorem.
- Squares numbers correctly.
- Gives answer with units.

2.

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = 12$$

Answer: 12 cm

Marking guidance:

- Identifies the hypotenuse.
- Rearranges correctly.
- Gives answer with units.

3.

$$10^2 = 6^2 + h^2$$

$$100 = 36 + h^2$$

$$h^2 = 64$$

$$h = 8$$

Answer: 8 m

Marking guidance:

- Forms the correct equation.
- Solves accurately.
- Includes units.

IMPORTED FROM CHAT:

To draft the worksheet, I need one detail: what is the topic (and year/stage) you want the worksheet to cover? For example: Year 9 Maths – Linear Relationships, Stage 5 Science – Chemical Reactions, or Year 11 Maths Advanced – Trigonometric Functions.

— [TEACHER ONLY] — Consider adding a misconception-check question where students must identify the hypotenuse before applying Pythagoras' theorem, as Year 8 students commonly square the wrong side.

ChatGPT Free: <https://chatgpt.com/share/6a3b1963-6838-83ec-85e5-eebf5448db33>

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Introduction to Pythagoras' Theorem

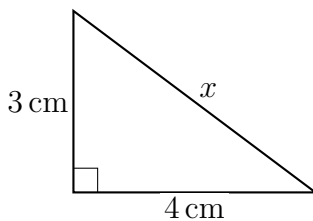
Learning Intention

To understand and apply Pythagoras' Theorem ($c^2 = a^2 + b^2$) to find missing side lengths in right-angled triangles. Remember that c must always be the **hypotenuse** (the longest side, directly opposite the right angle).

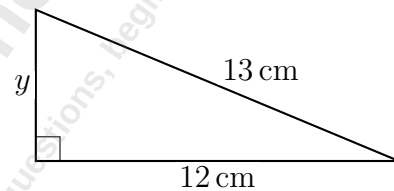
1. Fluency: Finding missing sides

For each of the right-angled triangles below, use Pythagoras' Theorem to calculate the value of the unknown side. Show all steps of your working.

a) Find the hypotenuse, x .



b) Find the shorter side, y .



2. Application: The Farm Gate

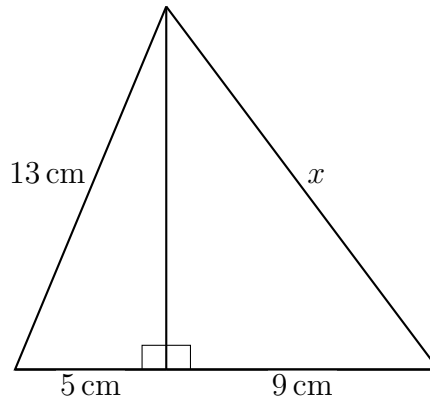
A farmer is building a rectangular wooden gate that is 2.4 m wide and 1.0 m high. To ensure the gate does not sag, she needs to attach a diagonal brace across the back.

- Draw a simple, labelled diagram to represent this scenario.
- Calculate the exact length of the diagonal piece of wood required.

3. Challenge: The Composite Shape

The diagram below shows two right-angled triangles joined together. Calculate the length of the side marked x .

Hint: You will need to find the length of the shared middle side first.



IMPORTED FROM CHAT:

I am ready to design your LaTeX worksheet. Could you please provide your teacher brief, including the subject (Maths, Science, or Integrated STEM), Year/Stage, topic, and preferred pedagogy mode?

— [TEACHER ONLY] — *Refinement Suggestion: If you notice students struggling with identifying whether to add or subtract squares, I can add a quick "Misconception Check" diagnostic question before the fluency task where they must spot and explain the error in a provided, flawed calculation.*

Gemini chat:

<https://gemini.google.com/share/d/1OEdOeqMA57jSFht7kKp3kr2ZtA5Pa0Lq?usp=sharing>

Teacher Answer Key

Question 1: Fluency

- **1a)** Using $c^2 = a^2 + b^2$

$$x^2 = 3^2 + 4^2$$

$$x^2 = 9 + 16$$

$$x^2 = 25$$

$$x = \sqrt{25} = 5 \text{ cm}$$

Marking guide: 1 mark for correct substitution, 1 mark for correct final answer with units.

- **1b)** Using $a^2 + b^2 = c^2$

$$y^2 + 12^2 = 13^2$$

$$y^2 + 144 = 169$$

$$y^2 = 169 - 144$$

$$y^2 = 25$$

$$y = \sqrt{25} = 5 \text{ cm}$$

Marking guide: 1 mark for identifying the hypotenuse is 13 and setting up the subtraction/equation correctly, 1 mark for final answer.

Question 2: Application

- **2a)** Students should draw a rectangle with base 2.4 m, height 1.0 m, and a diagonal drawn in. A right-angle symbol should be present in the corner.

- **2b)** Using $c^2 = a^2 + b^2$

$$c^2 = 2.4^2 + 1.0^2$$

$$c^2 = 5.76 + 1.0$$

$$c^2 = 6.76$$

$$c = \sqrt{6.76} = 2.6 \text{ m}$$

Marking guide: 1 mark for diagram, 1 mark for calculation, 1 mark for final answer in metres.

Question 3: Challenge

- **Step 1: Find the shared vertical side (let's call it h)**

Looking at the left triangle:

$$h^2 + 5^2 = 13^2$$

$$h^2 + 25 = 169$$

$$h^2 = 144$$

$$h = \sqrt{144} = 12 \text{ cm}$$

- **Step 2: Find x**

Looking at the right triangle:

$$x^2 = h^2 + 9^2$$

$$x^2 = 12^2 + 9^2$$

$$x^2 = 144 + 81$$

$$x^2 = 225$$

$$x = \sqrt{225} = 15 \text{ cm}$$

Marking guide: 1 mark for finding the intermediate side (12cm), 1 mark for setting up the second equation, 1 mark for correct final answer.

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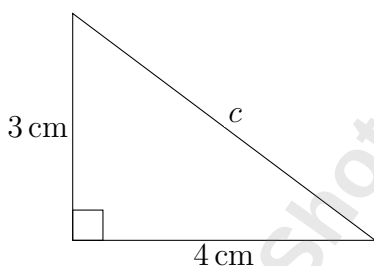
Introduction to Pythagoras' Theorem

Key idea: In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides:

$$a^2 + b^2 = c^2$$

1. Find the hypotenuse

A right-angled triangle has side lengths 3 cm and 4 cm.



Find the length of the hypotenuse.

2. Find a missing side

A right-angled triangle has hypotenuse 13 cm and one side 5 cm.

Find the length of the other side.

3. Challenge: Word problem

A ladder leans against a wall. The base of the ladder is 6 m from the wall, and the ladder is 10 m long.

- How high up the wall does the ladder reach?

Teacher Answer Key

1.

$$c^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$c = \sqrt{25} = 5 \text{ cm}$$

Answer: 5 cm

2.

$$c^2 = a^2 + b^2$$

$$13^2 = 5^2 + b^2$$

$$169 = 25 + b^2$$

$$b^2 = 144$$

$$b = \sqrt{144} = 12 \text{ cm}$$

Answer: 12 cm

3.

$$c^2 = a^2 + b^2$$

$$10^2 = 6^2 + h^2$$

$$100 = 36 + h^2$$

$$h^2 = 64$$

$$h = \sqrt{64} = 8 \text{ m}$$

Answer: 8 m

IMPORTED FROM CHAT:

I'm ready to build this, but I need a bit more detail first. What are the key inputs: subject (Maths/Science), year or stage, topic, number of questions, and desired difficulty or mode (e.g. exam-style, scaffolded, investigation)?

— [TEACHER ONLY] —

Add a misconception check (e.g. a triangle that is not right-angled) to ensure students do not apply Pythagoras indiscriminately.

Perplexity Chat:

<https://www.perplexity.ai/search/90e8d150-9962-463f-8268-ee5ea7ac3962>